A Comparative Study of Logistic Regression and Naïve Bayes for the Prediction of Students' Exam Performance

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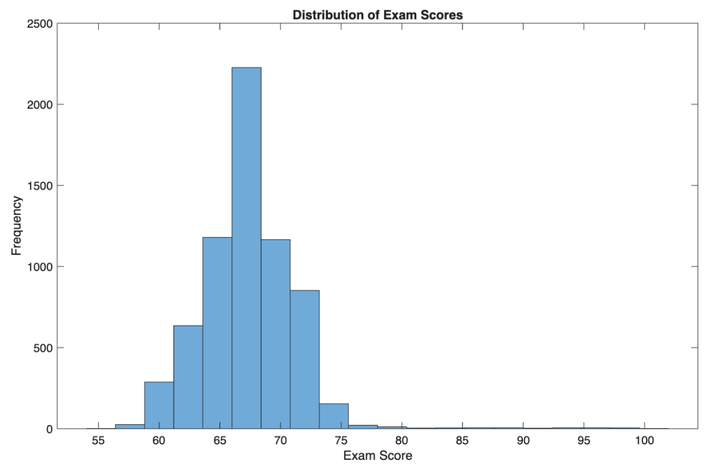
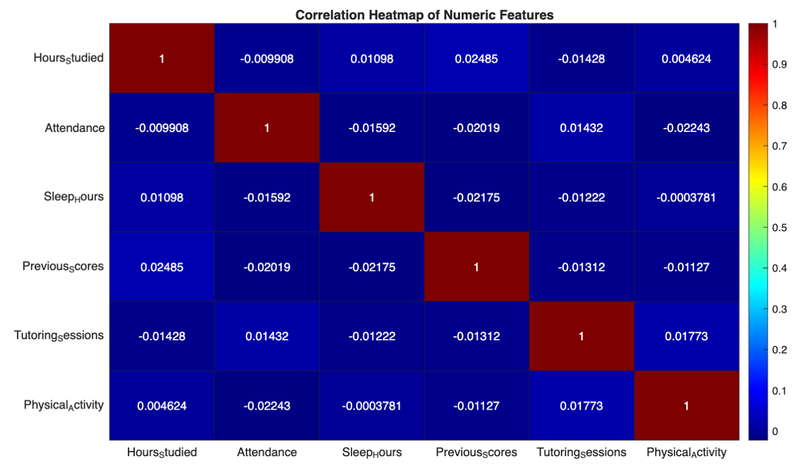
Introduction

- The student academic performance is influenced by multiple factors, such as behavior, education and school of type.
- To increase students' performance in academic and provide relevant assistance to them, it is necessary to predict the exam outcomes in the early stage.
- This study formulates the task as a binary classification problem to predict whether a student's exam score results in a PASS or FAIL on the basis of performance-related attributes.



Initial Analysis of the Dataset

- The dataset is from Kaggle, Student Performance Factors.
- Mosapc (2025) published a dataset on Kaggle that contains 6,607 records with 20 features, including both numeric and categorical variables.
- The target variable is derived from the exam score and converted into 0 and 1 binary outcome for classification.
- The initial data analysis shows meaningful variation across features and moderate correlation between academic-related variables. This suggests that the data is suitable for supervised learning.



The Machine Learning (ML) methods and their advantages and disadvantages

The Logistic Regression model

- Charizanos (2024) illustrated that the the Logistic Regression (LR) is widely applied for the binary outcome to the probabilistic classifier and allows regularisation machine learning technique to control the model complexity.
- Advantage and Disadvantage:
  - LR has stable decision boundaries due to regularisation, which adds a penalty to the model's complexity. However, it assumes linear separability and, thus, is sensitive to outliners.

Naïve Bayes model

- Naïve Bayes (NB) is another machine learning method for the probabilistic classifier based on Bayes' theory, assuming that all features are conditionally independent (Sinha *et al.*, 2017).
- Advantage and Disadvantage:
  - NB is fast and simple. It performs well on high dimensional data with high computational efficiency. However, its results may be negatively affected when features are not independent and correlated.

Hypothesis

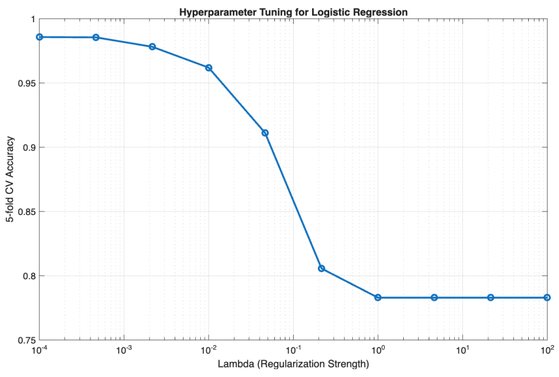
- Both methods are expected to perform better than random guessing. Due to correlated features and the use of regularisation, it is hypothesized that the logistic regression model will achieve superior performance than Naïve Bayes model in this machine learning model comparison.

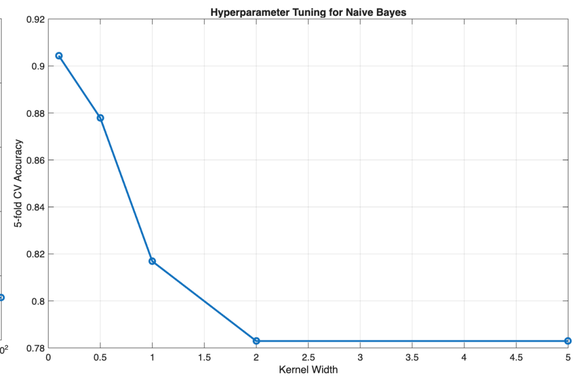
The Choice of Parameters and Experimental Setup

Training Strategy

- This dataset was split into training and test sets with 70/30 ratio with the aim to evaluate the generalisation of the performance (Pawluszek *et al*, 2020).
- To avoid data leakage, data preprocessing was applied consistently.

Cross Validation and Hyperparameter Tuning





- During model selection, 5-fold cross validation was used to reduce overfitting (Grimm, 2017).
- Logistic regression was regularised using an L2 penalty, with the optimal λ selected via cross-validated accuracy (Ward *et al.*, 2020).
- Naive Bayes was implemented with kernel density estimation, with the kernel width optimised using cross-validation (Zhang *et al.*,2018).

Evaluation Metrics

- Performance was evaluated using accuracy, precision, recall, F1-score, and ROC–AUC to provide a balanced and threshold-independent comparison.
- Accuracy alone may be misleading when misclassification costs differ. Therefore, F1-score and ROC–AUC were included to provide a more balanced and threshold-independent evaluation (Carrington *et al*,2020).
- In the context of students' performance prediction, recall is particularly important as false negatives correspond to students at risk who are incorrectly predicted to pass.
- ROC–AUC was used to assess model performance across all classification thresholds, providing a robust comparison independent of decision boundary selection.
- Consistent performance across multiple metrics increases confidence in the robustness of the selected model.

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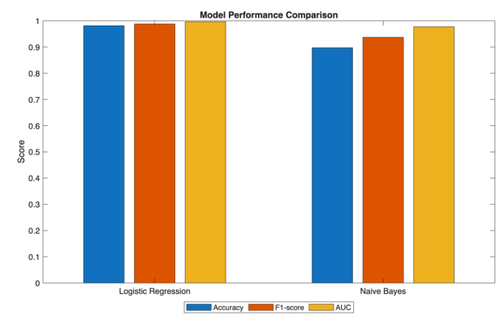
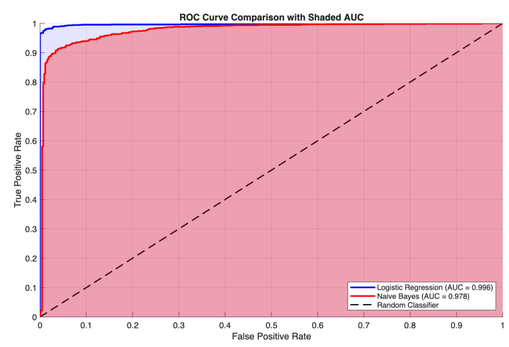
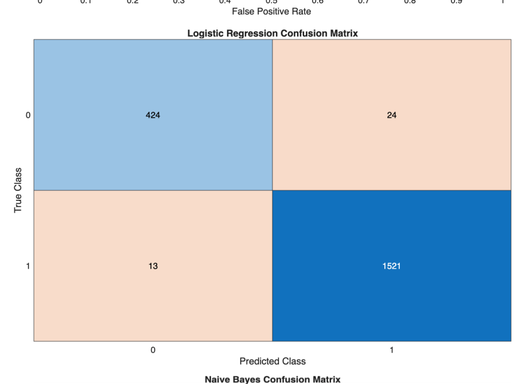
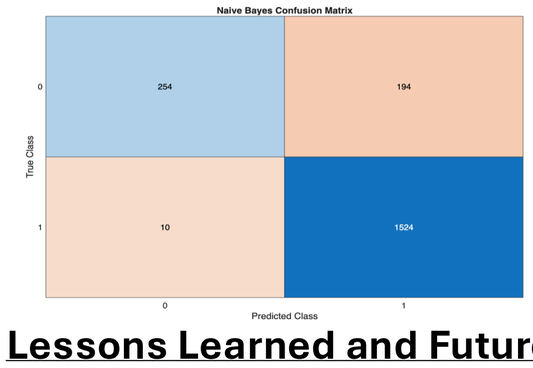
Result Analysis and Critical Evaluation

ML Methods Performance Comparison









- Both models perform substantially better than random guessing. However, the performance of Logistic Regression consistently outweighs Naive Bayes across all evaluation metrics, including F1-score and AUC.
- The reason is that the ability of Logistic Regression does not assume conditional independence between features, which is unlikely to apply to educational variables such as study time, attendance, and academic support.
- The use of L2 regularisation in Logistic Regression helps control model complexity (Huang *et al*, 2016) and reduces overfitting, which contributes to more stable generalisation performance.
- The ROC curve shows that Logistic Regression consistently achieves a higher true positive rate at comparable false positive rates, resulting in a larger AUC.
- Confusion matrix analysis indicates that Logistic Regression produces fewer false negatives, which is desirable in identifying students at risk of failing (Düntsch *et al*, 2019).
- Despite lower performance, Naive Bayes remains computationally efficient and may be suitable for scenarios with limited data or strong independence assumptions.
- Overall, Logistic Regression achieves a better bias–variance trade-off for this dataset, while Naive Bayes demonstrates higher bias due to its simplifying assumptions.

Limitation

- Cross–validation and an independent test set help mitigate overfitting, although the high AUC values suggests that the dataset may contain specific patterns. However, the results may vary when applied to different educational contexts.

Lessons Learned and Future Work

- This study highlights the importance of proper preprocessing of data handling, cross-validation and the use of multiple evaluation metrics when comparing classifiers.
- ROC Curve with AUC proved useful for threshold-independent evaluation.
- To further improve generalisation of the machine learning model, my future work would explore additional classifiers such as Random Forests and Support Vector Machines, address potential class imbalance and incorporate features.